S Raghav Pillai

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Objective

Platform/Infrastructure Engineer with 6+ years of experience building **internal frameworks, developer tooling, CI/CD systems, and scalable infrastructure**. Passionate about **reducing DevOps toil, improving reliability, and designing abstractions** that empower developers to move fast without breaking systems.

Education

Qualification	Institution	Year	Score
Bachelor of Technology *Computer Science and Engineering*	SRM Institute of Science And Technology, Chennai	2016-2020	CGPA: 8.13
Class XII *Central Board of Secondary Education*	St. Mary's Senior Secondary School, Delhi	2016	87.6%

Achievements

- **Common Vulnerabilities and Exposures**
  CVE ID 2019-13143, 1st July 2019
  View at: cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2019-13143

Experience

Moveworks India (Bangalore)
Senior Software Engineer 1, *March 2023 - Present*,

- **Architected and built the high-performance database layer** for Moveworks' enterprise search platform, managing **60+ TB** of multi-tenant ingested data while maintaining sub-**100 ms** search query latency.
- **Engineered a federated cross-region ingress gateway** to handle enterprise traffic across distributed data centers, enabling dynamic traffic re-routing and high availability.
- **Designed a fine-grained, field-level access control system** for OpenSearch by integrating **OpenFGA** and building a custom intercepting proxy to enforce dynamic authorization before returning query results.
- **Developed an automated infrastructure health-check framework** that dynamically inspects service configuration dependencies (PostgreSQL, Redis, Kafka) and executes end-to-end connection validation tests.
- **Built a Python-based Argo Workflows automation framework**, enabling developers to independently author, test, and deploy multi-environment pipelines—significantly reducing DevOps overhead and human error while establishing standardized deployment conventions.
- **Integrated Bazel into Kubernetes manifest repositories** to establish automated pre-merge checks, including dynamic manifest diffing and validation within CI/CD pipelines.
- **Managed core service mesh (Istio), Opensearch cluster, Logging infra, Kafka clusters**, **Config delivery systems**, **Data governance layer** etc.

Software Engineer 1, *April 2022 - March 2023,*

- Responsible for managing cloud infrastructure, building CI-CD pipelines and tooling.
- Migrated services from a monolithic deploy pipeline to service-specific deploy pipelines, effectively

bringing a service's deployment time from well over 2 hours to less than 10 mins.
- Set up infra in the EU region to meet GDPR guidelines.

Citrix R&D India (Bangalore)
Platform Engineer 2, *Nov 2021 - April 2022*

- Responsible for building and maintaining Internal services on Azure.
- Wrote automation to publish a list IP ranges to whitelist for customers using our product.

Zolostays (Bangalore)
Platform Engineer 2, *July 2020 - Oct 2021*,

- Developed internal services (ZOps), this service, helped developers manage configs and service deployments.
- Managed infrastructure, VPN, Users access, and Microservices,
- **Started** zolo's **bug bounty program**,
- Took measures to **secure** the **infrastructure**,
- Performed **cost optimisation** exercises and saved over **$2000 per month**,

Intern, *Jan 2020 - June 2020*

- Zolostays is an Indian startup which offers Co-living services in various cities. Here I spent my time building, deploying and managing the infrastructure of multiple microservices.

Technical Skills

- Kubernetes *Advanced*, Docker *Intermediate*, Linux *Intermediate*, Jenkins *Intermediate*
- GoLang *Intermediate*, Python *Advanced*, Rust *Familiar*
- AWS (*EC2, EKS, S3, Lambda, WAF, GuardDuty, Config, Cloudwatch etc)*
- Terraform *Intermediate*, Bazel *Intermediate*
- ProtoBuffers *Intermediate*, GRPC *Intermediate*,
- PostgresSQL *Intermediate*, MongoDB *Familiar*, ElasticSearch *Intermediate*, Kafka *Intermediate*, Clickhouse *Intermediate*

Projects

- **ArgoWorkflow Framework**
  An abstraction layer on top of python hera framework for argo workflows that allow developers to write cross environment workflows without any DevOps involvement.
- **Manifest generation framework**
  A Bazel rule backed by a python framework that takes params like resource requirements as inputs and generates the necessary k8s manifests to deploy a microservice in AWS EKS.
- **zOps**
  A project to create and manage Configmaps and secrets for various applications through a web console. Integrated with the proprietary access control system (Bifrost) of Zolostays.
- **Ragnarok (Cluster Shutdown)**
  A serverless project to turn off/scale down all staging clusters during nights and weekends which can also be turned on at the push of a button during the shutdown hours.
- **PinNinja**
  github.com/securelayer7/PinNinja
  An Arduino script to identify UART pins.
- **Rubber Ducky Using an Arduino Leonardo Pro Micro**
  github.com/vologue/Rubber-ducky-using-arduino-leonardo-pro-micro
  An Arduino script that provides the attacker with a shell from the victim's machine.

Activities & Involvements

- Presenter at PyCon India 2019.